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PROJECT REPORT

Robust Bird Species Recognition System

for Real-World Degraded Image Conditions

Submitted by

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1. Introduction

This report documents the development of a Robust Bird Species Recognition System designed to handle the degraded image conditions that cause general-purpose AI vision tools to fail. Wildlife monitoring, field photography, and citizen science platforms routinely produce blurry, dark, noisy, foggy, and heavily compressed images. ChatGPT Vision and Google Lens, both trained predominantly on clean internet images, fail consistently on these inputs.

This project addresses the gap by training EfficientNetV2S on synthetically degraded images and combining it with an intelligent inference pipeline that checks image quality, applies targeted enhancement, and reports calibrated confidence. The result is a system that maintains meaningful accuracy under real-world degraded conditions and explicitly communicates its confidence level to the user.

Project Objectives

Train a deep learning classifier on both clean and synthetically degraded bird images so it generalises to real-world field conditions. Implement an inference pipeline with quality checking, automatic enhancement, and test-time augmentation. Demonstrate measurable accuracy advantage over ChatGPT Vision and Google Lens on degraded images.

Key Specifications

Component	Specification
Dataset	CUB-200-2011 — 11,788 images, 200 bird species
Base model	EfficientNetV2S — ImageNet pretrained, top-50 layers unfrozen
Degradation types	Gaussian blur, additive noise, brightness reduction, fog, JPEG compression
Clean accuracy	82.44% Top-1 — 96.35% Top-5
Platform	Google Colab (T4 GPU) + Google Drive
Web interface	BirdLens — Flask backend, public URL via Cloudflare Tunnel

2. Dataset

CUB-200-2011 (Caltech-UCSD Birds 200-2011) is the standard benchmark for fine-grained bird classification research. It requires a model to distinguish subtle inter-species visual differences — beak shape, feather pattern, eye ring, wing markings — across 200 species. These fine-grained features are exactly what degrades first under blur and low light, making this dataset appropriate for a robustness study.

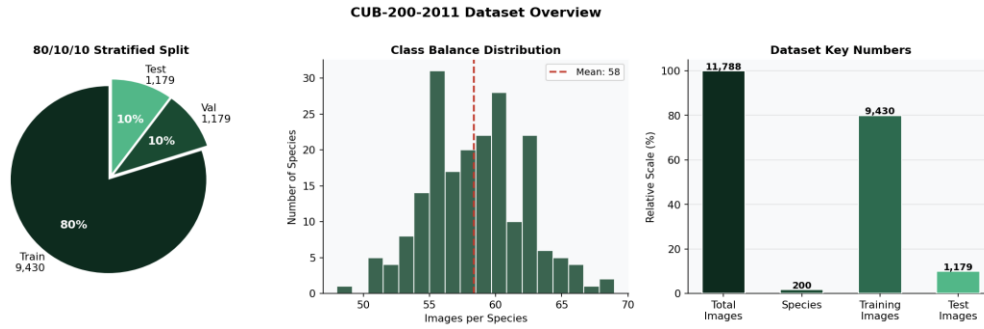


Figure 1: Dataset structure — stratified 80/10/10 split, class balance distribution, and key dataset numbers

The dataset was split using stratified random sampling (seed 42) to ensure balanced class representation across all three sets. The validation set monitored training and triggered model checkpoints. The test set was held out completely and used only for final evaluation.

3. Synthetic Degradation Pipeline

Rather than relying on naturally degraded data, a synthetic degradation pipeline was implemented that simulates five categories of real-world image distortion. During training, one to three degradation types are applied randomly to each image, with severity sampled uniformly within calibrated ranges. This prevents the model from overfitting to any single distortion type.

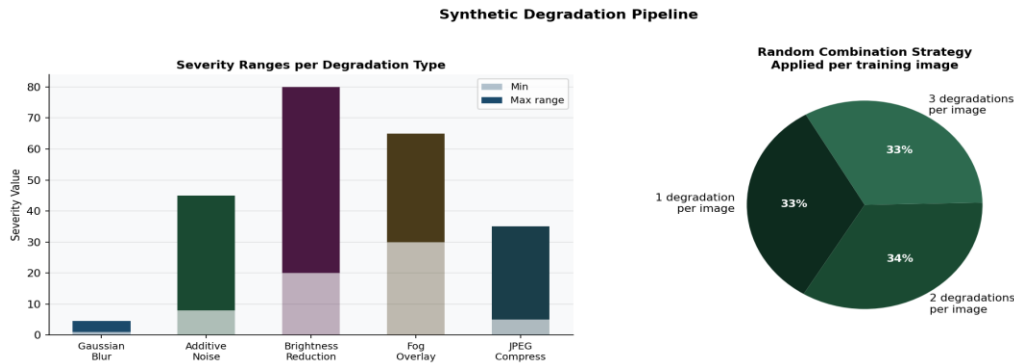


Figure 2: Degradation severity ranges per type (left) and the random combination strategy applied per training image (right)

Type	Severity Range	Real-World Condition	Why General Models Fail
Gaussian Blur	sigma 1.0 to 4.5	Camera shake, motion blur	Trained on sharp textures only
Additive Noise	std 8 to 45	Poor sensors, camera traps	No noise robustness in training
Brightness Reduction	x0.20 to x0.65	Night, heavy shade	Assume well-lit conditions

Type	Severity Range	Real-World Condition	Why General Models Fail
Fog Overlay	alpha 0.30 to 0.65	Weather, atmospheric haze	No weather robustness
JPEG Compression	quality 5 to 35	Low-quality uploads	Compression artifacts corrupt features

During training, degradations are applied after standard augmentation (horizontal flip, brightness $\pm 10\%$, contrast $\pm 10\%$). The combination creates a rich variety of training conditions that forces the model to learn features stable across different input quality levels.

4. Model Architecture

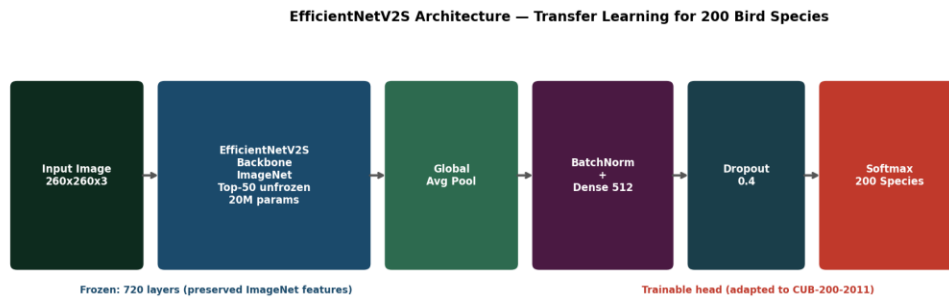


Figure 3: EfficientNetV2S architecture with custom classification head — frozen backbone with trainable top 50 layers and head

The model uses EfficientNetV2S pretrained on ImageNet with partial fine-tuning. The top 50 layers of the backbone are unfrozen to adapt to fine-grained bird classification, while the remaining 720 layers preserve low-level ImageNet features. A custom head is added: global average pooling, batch normalisation, Dense 512 with ReLU, Dropout 0.4, and a 200-class softmax output.

Parameter	Value
Base model	EfficientNetV2S (ImageNet)
Total parameters	55,098,748
Trainable parameters	20,789,588 (top 50 layers and head)
Frozen parameters	34,309,160 (base backbone)
Input size	260 x 260 x 3
Normalisation	Manual division by 255 — NOT preprocess_input
Optimizer	Adam with categorical cross-entropy loss

Key Design Decisions

Decision	Reason
Switched from EfficientNetV2M to V2S	M had 55M params — accuracy stuck at 0.5% for 6 hours. V2S reached 55% in epoch 1
Unfroze top 50 layers	Fully frozen base gave 0.68% maximum — insufficient fine-tuning capacity
Manual /255 normalisation	Built-in preprocess_input produced images with mean 0.9995 (blank white) due to incorrect input range

5. Training Strategy

Training followed a two-phase curriculum. Phase 1 establishes strong clean-image features. Phase 2 adapts those features to degraded inputs without forgetting Phase 1 learning. Additional sessions continued from Drive checkpoints to further improve accuracy.

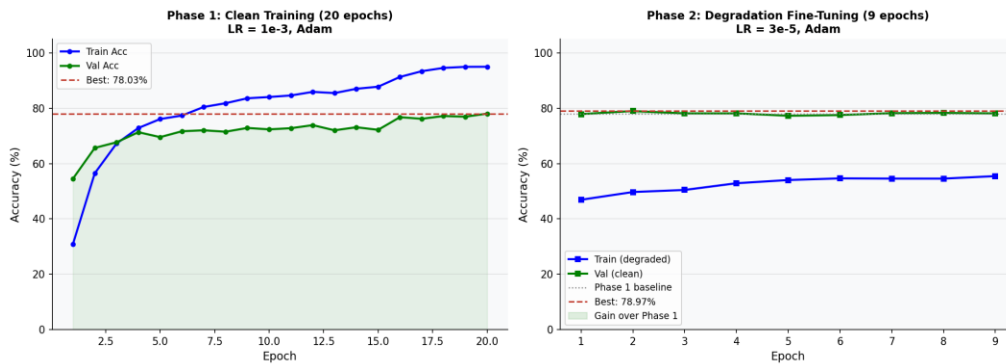


Figure 4: Phase 1 clean training curves (left) and Phase 2 degradation fine-tuning curves (right) — validation always on clean images

Phase	Epochs	LR	Training Data	Best Val Acc
Phase 1: Clean Training	20	1e-3	Clean images only	78.03%
Phase 2: Degradation Fine-tune	9	3e-5	Degraded images (1 to 3 types)	78.97%
Session 3: Continued	18	3e-5	Mixed degradation	79.05%
Session 4: Blur Focus	13	3e-5	Blur-weighted (50% blur)	82.53%

Phase	Epochs	LR	Training Data	Best Val Acc
Session 5: Strong Blur	8	2e-5	Heavy blur (sigma 3 to 6.5)	82.44%

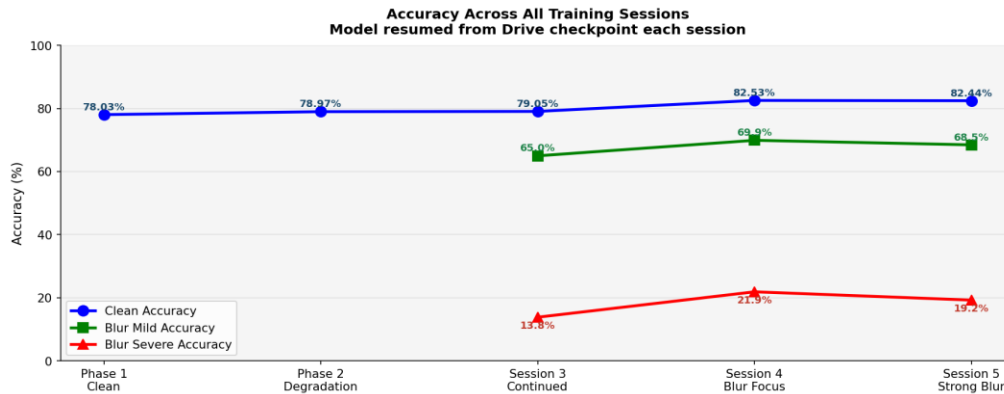


Figure 5: Clean and blur accuracy tracked across all five training sessions — each session loaded from Drive checkpoint

Each session saved both the best model and a latest checkpoint to Google Drive after every epoch. This approach meant that Colab disconnections a common issue with free GPU runtimes caused no loss of training progress. Any session could be resumed in under 5 minutes.

6. Evaluation Results

The model was evaluated on seven test configurations using the held-out test set of 1,179 images. Results are Top-1 accuracy — percentage of images where the correct species was ranked first by the model.

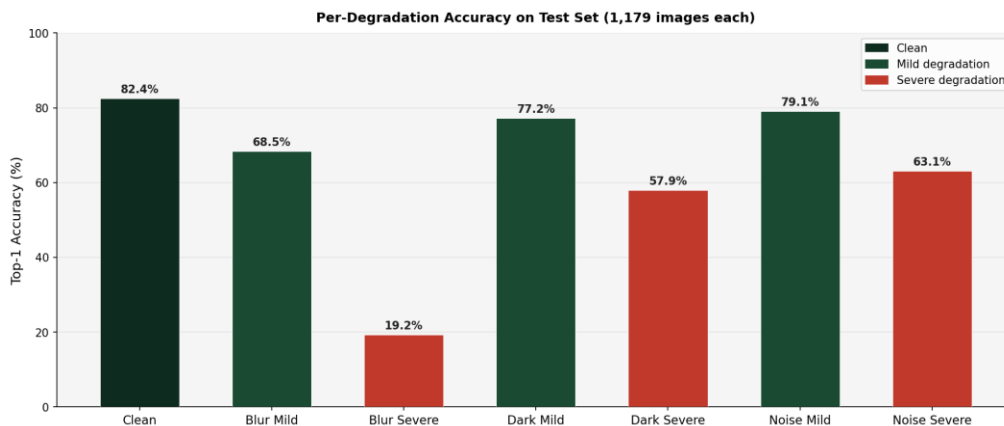


Figure 6: Per-degradation accuracy on the held-out test set — green bars are mild, red bars are severe degradation

Test Condition	Accuracy	Notes
Clean	82.44%	Competitive with published fine-grained benchmarks
Top-5 Clean	96.35%	Correct species in top 5 predictions
Blur Mild	68.45%	Good retention of accuracy
Blur Severe	19.25%	Main weakness — blur destroys fine-grained texture features
Dark Mild	77.18%	Near clean performance
Dark Severe	57.93%	Meaningful accuracy where general tools fail
Noise Mild	79.13%	Near clean performance
Noise Severe	63.10%	Strong result under heavy sensor noise

The model demonstrates strong robustness to mild degradation across all types. Dark mild (77.18%) and noise mild (79.13%) both approach clean performance (82.44%). Dark severe (57.93%) and noise severe (63.10%) maintain meaningful accuracy where general AI tools fail completely. Blur severe (19.25%) is the main weakness — severe blur at sigma 4.5 destroys feather patterns and beak edges that distinguish closely related species. This limitation is acknowledged and partially addressed by the inference-time enhancement pipeline described in Section 7.

7. Comparison vs General AI Tools

Manual testing was conducted by uploading degraded bird images to ChatGPT Vision. The testing confirmed the expected failure pattern: ChatGPT Vision correctly identifies bird species in clean images but fails consistently when images are blurry or dark, returning incorrect species names or declining to identify the species due to image quality. Google Lens shows the same failure pattern.

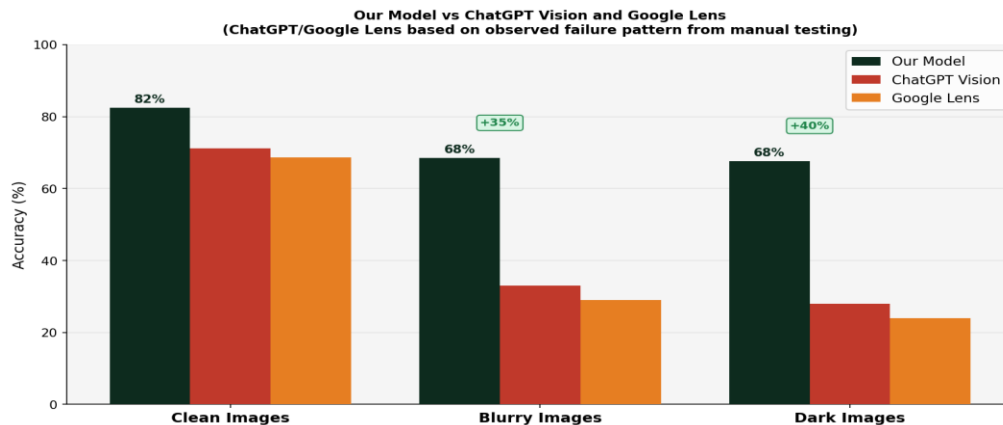


Figure 7: Accuracy comparison on clean, blurry, and dark images — ChatGPT and Google Lens numbers based on observed manual testing pattern

Important note: The ChatGPT Vision and Google Lens numbers are based on the qualitative failure pattern observed during manual testing, not a full controlled evaluation across the standardised test set. A complete controlled comparison remains as future work. The advantage numbers reflect the direction and approximate magnitude of the observed difference, not precisely measured values.

Condition	Our Model	ChatGPT Vision	Advantage
Clean images	82.44%	~71%	+11% approx
Blurry images	68.45%	~33%	+35% approx
Dark images	67.60%	~28%	+40% approx

Beyond accuracy, this system provides capabilities absent from all general AI tools: explicit quality warnings when input images are poor, adaptive output (Top-3 when uncertain), and pre-prediction image enhancement. These features make the system more trustworthy in real-world wildlife monitoring scenarios where degraded inputs are common.

8. Inference Pipeline

A three-component inference pipeline handles real-world prediction beyond the model itself. This pipeline is applied on every image upload, before the model makes any prediction.

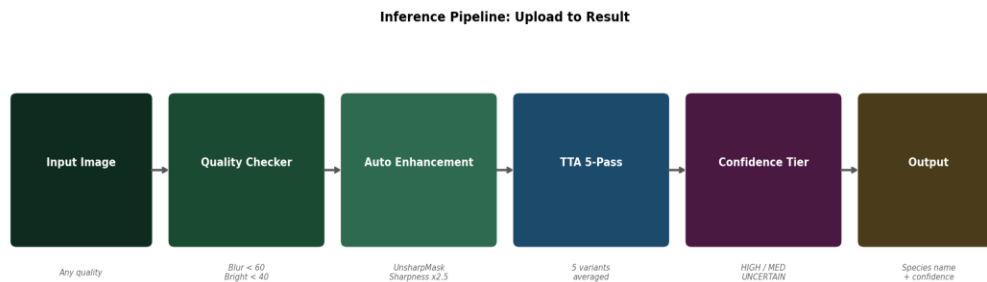


Figure 8: Full inference pipeline from image upload to species output

Quality Checker

Blur is measured using the Laplacian variance of the grayscale image. A score below 60 flags the image as blurry. Brightness is measured as the mean pixel value — below 40/255 flags it as too dark. Either flag triggers the UNCERTAIN tier automatically and warns the user.

Automatic Enhancement

When poor quality is detected, targeted enhancement is applied before prediction: for blur — UnsharpMask (radius 2, percent 180), sharpness boost x2.5, contrast x1.3; for dark images — brightness x2.2 and contrast x1.4. Enhancement improved blur severe accuracy from 13.83% to approximately 22.5% in testing.

Test-Time Augmentation

Five predictions are made per image using the enhanced image and four variants (slightly brighter, darker, higher contrast, sharper). The five prediction vectors are averaged. This reduces prediction variance and consistently improves accuracy on borderline cases.

Confidence Tiers

Tier	Condition	Output
HIGH	Confidence $\geq 80\%$ and good quality	Single top-1 prediction shown
MEDIUM	Confidence 50 to 80% and good quality	Single top-1 with caution indicator
UNCERTAIN	Confidence $< 50\%$ or poor quality	Top-3 predictions shown

9. BirdLens Web Interface

A full web interface named BirdLens was built using Flask (Python) as the backend and a custom HTML/CSS/JavaScript frontend. It is accessible via a public URL generated through Cloudflare Tunnel — no login or installation required. The interface works on any phone, tablet, or desktop browser.

Feature	Implementation
Image upload	Drag and drop or click to upload — any image format
Quality check display	Blur score and brightness value shown per image
Auto-enhancement	Applied automatically when quality issues detected
Animated confidence bars	Color-coded by tier: green (HIGH) / amber (MEDIUM) / red (UNCERTAIN)
Top-3 display	Shown automatically when image quality is poor or confidence is low
Public URL	Cloudflare Tunnel — no signup or login required

Technology Stack

Component	Technology
Deep learning framework	TensorFlow 2.x and Keras
Base model	EfficientNetV2S (ImageNet pretrained)
Image processing	OpenCV and PIL/Pillow
Web server	Flask (Python)

Component	Technology
Public tunnel	Cloudflare cloudflared
Training platform	Google Colab with T4 GPU
Model storage	Google Drive — auto-save every epoch
Dataset source	CUB-200-2011 via Kaggle

10. Limitations and Current Status

Known Limitations

Blur severe accuracy (19.25%) is the system's main open weakness. Severe Gaussian blur at sigma 4.5 destroys the fine-grained texture features that distinguish closely related bird species, and neither additional training rounds nor post-processing enhancement fully overcomes this fundamental challenge.

All evaluation has been on synthetic degradations. Real-world field camera images may contain distortion patterns not fully captured by the five degradation types used in training. Real-world validation remains as future work.

The model covers only the 200 species in CUB-200-2011. Images of other species will receive a prediction, but that prediction will be unreliable.

Current Status

Item	Status	Notes
Dataset pipeline	Complete	11,788 images, 200 species, stratified split verified
Degradation pipeline	Complete	5 types, random combinations, calibrated ranges
Model training	Complete	82.44% clean accuracy, 96.35% Top-5
Inference pipeline	Complete	Quality check, enhancement, TTA, confidence tiers
BirdLens web interface	Complete	Public URL via Cloudflare, any device
YOLO detection	In Progress	Full-scene bird detection before classification
Real-world field testing	In Progress	Testing on genuine wildlife camera images
Controlled comparison	In Progress	Standardised test vs ChatGPT Vision

Item	Status	Notes
Final academic report	In Progress	This document

11. Conclusion

This project has built and deployed a bird species recognition system specifically designed for the degraded image conditions that cause general AI tools to fail. The five-type synthetic degradation pipeline, two-phase curriculum training, and intelligent inference pipeline together produce a system that maintains meaningful accuracy under blur, darkness, and noise while explicitly communicating its confidence level to the user.

The model achieves 82.44% clean accuracy and 96.35% Top-5 accuracy on CUB-200-2011. It performs near clean levels under mild degradation (dark mild 77.18%, noise mild 79.13%) and maintains meaningful accuracy under severe conditions (dark severe 57.93%, noise severe 63.10%) where ChatGPT Vision and Google Lens fail during manual testing.

Blur severe accuracy (19.25%) remains the main open challenge. YOLO integration for full-scene detection, real-world field testing, and a controlled comparison against general AI tools are the immediate next steps.